Stage 5 - Check Your Understanding

Q1. Bootstrap p-value

The bootstrap p-value is p_boot = (1 + number of simulated likelihood-ratio statistics at least as large as the observed statistic)/(B + 1). For mpox, Table 10 gives observed Lambda = 126.84 (approximately 126.8), B = 60, and zero exceedances.

Calculation: p_boot = (1 + 0)/(60 + 1) = 1/61 = 0.0163934, or approximately 0.0164 < 0.02.

This is the smallest attainable bootstrap p-value with 60 replicates because of the plus-one correction. To resolve smaller p-values, increase B: with 1,000 replicates and zero exceedances the minimum is 1/1001 = 0.000999. More simulations do not guarantee zero exceedances.

The asymptotic p-value (< 0.001) and bootstrap p-value both test the ordinary-model null alpha = 1; the former uses a theoretical reference distribution and the latter a simulated null distribution.

Q2. Why the AHM order is invisible

Proposition 12 states that the AHM-SEIR model preserves mean infectious period 1/gamma, reproduction number R0 = beta/gamma, and final epidemic size for every alpha. Only the distribution, especially the spread, of infectious periods changes.

As alpha tends to zero, for positive t the survival function tends to Phi(t) = (1/2) exp(-gamma t/2). Approximately half recover almost immediately, and the remaining half have an exponential duration with mean 2/gamma; the overall mean stays 1/gamma.

For mpox, the infectious period averages 2.9 days, shorter than a seven-day reporting interval, and the latent period averages 5.6 days. Weekly aggregation and the longer latent stage obscure changes in infectious-period spread. Figure 14 confirms near-flat AHM-SEIR profiles: log-likelihood changes by at most 0.6 over alpha from 0.4 to 1.

Q3. Memory versus misspecification for mpox

The pseudo-true alpha* is the order maximizing the expected likelihood of the fitted Caputo model when another process generates the observations. It is a property of the data-generating process and fitted model, not the random error in one data set.

The heterogeneous-recovery model yields alpha* = 0.636, versus the observed Caputo estimate 0.635: absolute difference = 0.001. Table 11 reports log-likelihood improvements over ordinary SEIR of 61.7 for heterogeneous recovery and 63.4 for Caputo, a difference of 1.7, with differing parameter counts.

Does mpox 2022 show memory? The mpox series strongly favors a more flexible description than ordinary SEIR, but it does not prove genuine fractional memory. A memoryless heterogeneous-recovery model produces almost the same fitted order and nearly the same likelihood improvement. Its fitted long-infectious minority should be treated as a statistical explanation, not as established mpox biology. The mechanistic AHM-SEIR order is not identifiable from the weekly series. Thus the current result is statistical equivalence,

with formal model comparison and forecasting deferred to Stage 6.

Q4. Fragility of the order

Figure 16 reports Liberia alpha-hat falling from 0.76 to 0.54 after flagged spikes are masked, while Lambda rises from 0.59 to 8.68. A single reporting spike therefore substantially affects the apparent evidence for memory.

For West Texas, the full-data alpha-hat is 1.00, but stopping four weeks after the peak produces alpha-hat = 0.40 (Lambda = 3.62). Section 6.2 explains that Caputo relaxation for alpha < 1 has a power-law tail: E_alpha(-lambda t^alpha) is asymptotically t^(-alpha)/[lambda Gamma(1-alpha)], not exponentially small.

West Texas has nine weeks with zero or one case among its final fourteen weeks. These observations penalize the persistent predicted tail for every alpha < 1. Truncating the data removes this penalty and permits a much smaller fitted alpha.

A parameter representing a fixed biological transmission property should be reasonably stable to a reporting spike or a modest change in observation window, apart from expected sampling uncertainty. The observed large shifts instead indicate that the fitted order is sensitive to curve shape and data coverage.